Progress Update 1 - Satellite Challenge

Mission Selection
217 - Ignis-1
Mission ID
217

Team Organization (ORG)
Mission Organizational Structure
Mission Leader;Karla Gabriela Morales Aguilar;a01642979@tec.mx;523320219098
Safety Manager;Iliana Cosette Saldaña Haro;a01648021@tec.mx;52332273855
Operational Manager;Barbara Ramos Aguilar;a01642652@tec.mx;523317895158
Technological Manager;Emiliano Rosario Martínez;a01642821@tec.mx;523331488792
Mission Name
IGNIS-1
Does your mission have any pending administrative documents to be submitted, such as a signed institutional letter?
No

Mission Organization (ORG) Pending Documents – For Missions with Actions Pending
Institutional Participation Letter
Already uploaded at the Entry Application Form
https://drive.google.com/file/d/1PV1ZGB4UwxfHmeGU_Z2vPXZeGfK9W9WI/view?usp=drive_link
Additional Updates
All the documentation has been completed and submitted at the Entry Application Form.

Team Demographics and Data Mapping
What is the main goal of this mission?
The main goal of our team is to design, develop, and successfully deploy a CanSat system capable of detecting early environmental conditions associated with forest fires while simultaneously evaluating wireless communication performance during descent. Through this mission, the team aims to apply engineering principles, integrate multiple technologies, and generate meaningful data such as a fire risk index and a communication quality index. This project also seeks to strengthen teamwork, technical skills, and innovation, aligning with the team's commitment to developing impactful engineering solutions.
Describe your team's strategy for developing, building, and presenting this mission at the Space Challenge.
The team will follow a structured and collaborative engineering approach consisting of design, prototyping, testing, and iterative refinement. Responsibilities are distributed among team members according to technical areas such as hardware integration, communication systems, and data analysis.
The development process is divided into three main phases: hardware integration, communication system implementation, and full system integration. The team will conduct ground testing to validate sensor performance, communication reliability, and system stability before flight. During the mission, data will be collected and later analyzed to evaluate performance and achieve mission objectives.

Throughout the process, the team will maintain clear communication, documentation, and coordination to ensure efficient workflow and successful presentation of results at the Space Challenge.

What is the total number of members in your team?

25 (4)

How many team members are under 18 years old?

0

How many team members have completed a Bachelor's degree?

1

How many team members hold a Master's degree?

1

How many team members hold a PhD?

1

Satellite Challenge

Describe the objective and goals of your Satellite Mission

The objective of this mission is to develop a CanSat system capable of detecting early environmental conditions associated with forest fires while simultaneously evaluating wireless communication performance during descent. The mission aims to collect and analyze environmental and communication data to generate meaningful indicators that support decision-making in emergency scenarios.

The specific goals of the mission include:

• Measuring key environmental variables such as temperature, humidity, pressure, gas concentration, and particulate matter.

• Evaluating wireless communication performance through signal strength (RSSI) and packet loss at different altitudes.

• Acquiring data during descent to analyze the relationship between environmental conditions and communication quality.

Regarding participation in the event on site, is your team willing to participate in the Sugarcane Launch Range?

Yes

Is this Satellite Mission linked to an accepted Rocket Mission?

Yes

If YES in the previous question, please inform the Rocket Mission ID

22

If NO in the previous question, do you want LASC to find an experimental rocket compatible with this mission?

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General (GEN) and Specific Requirements (CUB, CAN, PQB)

Satellite Mass

300

Satellite Dimensions

1C

Projected Satellite Budget

800

Briefly Describe the Satellite Systems and Components

The CanSat is designed as a multi-system platform integrating environmental sensing, communication, data acquisition, and power subsystems.
The flight computer is based on an ESP32 microcontroller, which manages data collection, processing, and communication. Environmental sensing is performed using a BME280 sensor for temperature, humidity, and pressure; a gas sensor (MQ-2 or MQ-135) for gas detection; and a particulate matter sensor for smoke detection.
The communication subsystem utilizes a LoRa module to enable long-range wireless data transmission and signal strength monitoring (RSSI).
Data storage is handled through a microSD module, allowing onboard recording of all measurements. The system is powered by a rechargeable LiPo battery, ensuring stable energy supply throughout the mission.
The internal structure is organized in layers, with sensors positioned externally for accurate measurements, electronics centrally located for protection and integration, and the power system placed at the base to ensure stability during descent.

Satellite Structure and Materials (SSM)

Describe the Selection of Materials and Tests for the Satellite Structure

The CanSat structure will be designed using lightweight and resistant materials such as PLA components and aluminum or acrylic for structural support. These materials are selected to ensure low weight, mechanical strength, and ease of fabrication while complying with size and mass constraints.
The internal structure will be organized to protect sensitive electronic components while allowing proper airflow for environmental sensors to ensure accurate measurements. Structural stability will also be considered to maintain balance during descent under parachute deployment.
Planned tests include mechanical integrity tests to verify resistance to vibrations and impact during launch and landing, as well as fit and assembly tests to ensure proper integration of all components. Additionally, functional tests will be conducted to validate sensor exposure, data acquisition, and system stability. These tests will ensure that the CanSat operates reliably under expected mission conditions.

Electronic Power System and Communications (ECS)

Does this Satellite have Solar Panels?

No

What is the Satellite's Battery Endurance?

4 hours

Does this Satellite have RBF Pins that Cut All Power from the Battery to the Satellite Systems?

Yes

Does this Satellite Incorporate Battery Circuit Protection for Battery Charging/Discharging?

Yes

Describe the Electronic Power Systems and Critical Wiring

Yes

Describe the Communications and Radio Frequency of the Satellite

The communication system is based on a LoRa (Long Range) module, which enables low-power, long-distance wireless transmission between the CanSat and the ground station.

The system operates in the ISM frequency band (typically 915 MHz or 433 MHz depending on regional regulations).
The communication architecture allows real-time transmission of environmental and system data, including temperature, humidity, pressure, gas concentration, particulate matter, and signal strength (RSSI). Additionally, packet loss will be monitored to evaluate communication reliability at different altitudes during descent.

Recovery System (REC)

Is this Satellite Mission designed to be recovered independently of the launch vehicle?

Yes, the satellite mission is designed to be recovered independently of the launch vehicle. After reaching the target altitude and being ejected from the rocket, the CanSat deploys a passive recovery system that allows a controlled descent. This ensures safe landing and full retrieval of the payload for post-flight data analysis.

Describe the Recovery System and Parachute Design of your Satellite

The recovery system of the CanSat is based on a passive parachute deployment mechanism that activates immediately after separation from the launch vehicle.
Once the satellite is ejected at the target altitude, a lightweight parachute is deployed to reduce the descent velocity and ensure a stable, controlled fall. The parachute is designed to maintain vertical orientation, which is critical for consistent environmental data collection during descent.

Agreement on LASC Terms and Conditions

Did you read and review the LASC Documentation System and understand the LSARP concept established in Section 2.3 of the LASC General Rules (LGR)?

Yes, I have read and reviewed the LASC documentation system, and I understand the LSARP concept

Do you accept the policies, procedures, and processes of the LASC Documentation System?

Yes, I hereby accept and agree to all LASC policies, procedures, and processes

Do you accept the Latin American Space Challenge Safety Policy?

Yes, I do accept the LASC safety policy